SECTION EN

Engine System

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General Instructions

WARNING

Detailed information on standard workshop and safety procedures, and general servicing operations is not included in this manual, which has been prepared to assist qualified service personnel. ARGO assumes no responsibility or liability for PERSONAL INJURY or VEHICLE DAMAGE which results from any servicing procedure performed, including those instructions outlined in this manual. Before performing a servicing operation, an individual must have determined to his/her satisfaction that personal injury or vehicle damage will not result from the servicing procedure or tools selected.

NOTE

All engine service work should be performed by an authorized Briggs and Stratton or Kawasaki mechanic. Severe damage and/or reduced performance can result from an improperly serviced engine.

CURRENT ENGINES USED IN VEHICLE PRODUCTION

⚠ IMPORTANT

For detailed servicing repairs, diagnostics and troubleshooting of these engines, please refer to the manufacturer's full service guide.



Briggs 700 EFI



Briggs 650



Briggs 600



Briggs 950 EFI



Briggs 850 EFI



Kawasaki 800 EFI



ODG 850



ODG 650

To check the oil during an operating period, shut the engine off, let it cool down and allow the oil time to drain into the sump before checking the oil level. Position the vehicle so the engine is level.

Check the engine oil level each day before operating the engine.

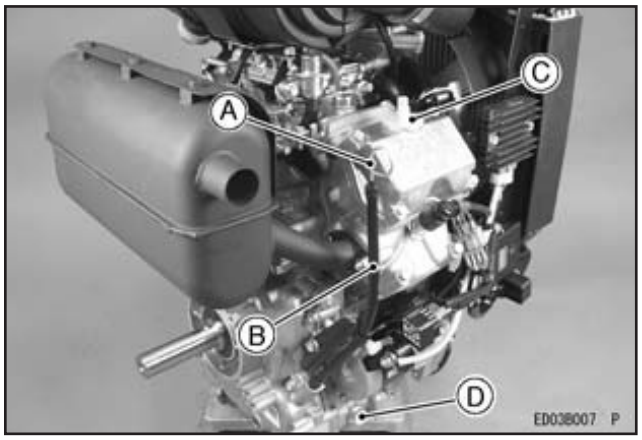
Briggs & Kawasaki Engines are equipped with a dipstick and a separate oil filler tube. To check the oil level, clean the area around the dipstick before removing. Remove the dipstick and wipe it with a clean cloth. Re-insert the dipstick and push it all the way into the tube. Remove the dipstick and check the oil level. If the level has dropped any, add oil to bring the level up to the FULL mark. DO NOT OVERFILL.



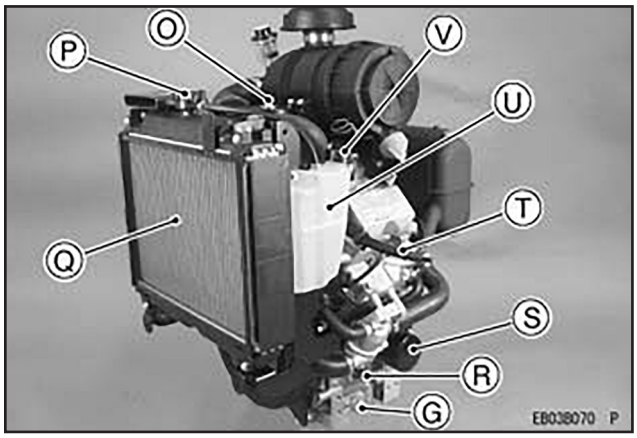
CAUTION

Do not run the engine if the oil level is above the FULL mark or below the ADD mark. Premature engine damage or total engine failure can occur when the oil level is not properly maintained.

Recommended Engine Oil Kawasaki 800



A. Dipstick
B. Dipstick Filler Tube
C. Oil Filler Cap
D. Oil Drain Plug

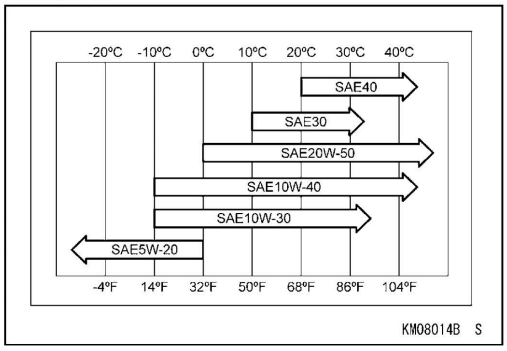


O. Throttle Body
P. Radiator Cap
Q. Radiator
R. Coolant Drain Plug
S. Oil Filter
T. Fuel Tube
U. Coolant Overflow Reservoir
V. Coolant OverflowReservoir Cap

Oil Type/Capacity Kawasaki EFI

Kawasaki Oil Viscosity Chart

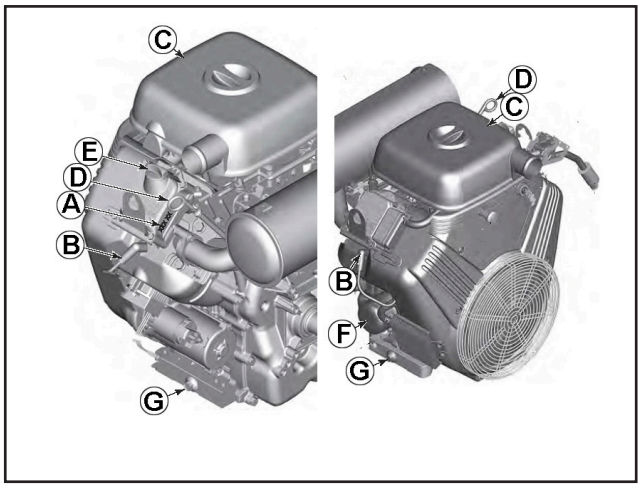
Choose the viscosity according to the temperature as follows:

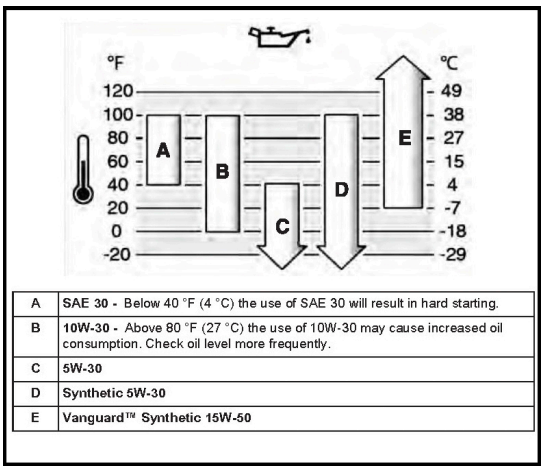


Engine Oil Capacity

FD851D	1.8 L (1.9 US·qt) [when oil filter is not removed]
	2.0 L (2.1 US·qt) [when oil filter is removed]

Briggs & Stratton Big Block EFI 850/950 - Oil Type/Capacity Briggs: 2.3 L (2.4 qts.)

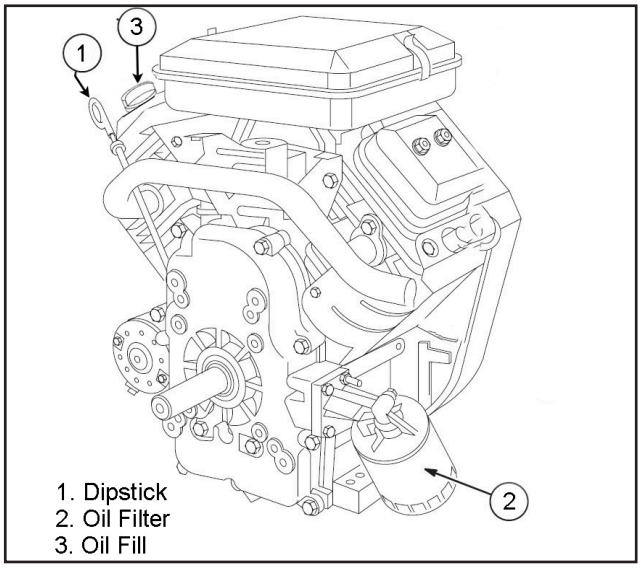


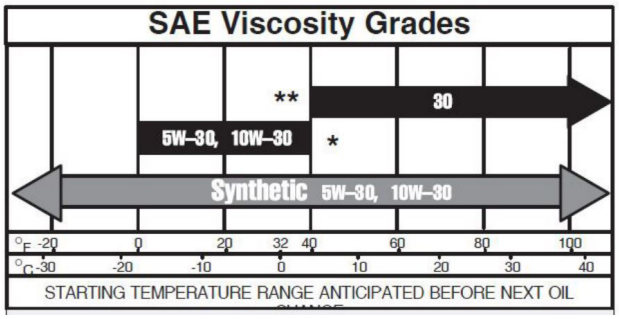


A	**SAE 30 -** Below 40 °F (4 °C) the use of SAE 30 will result in hard starting.
B	**10W-30 -** Above 80 °F (27 °C) the use of 10W-30 may cause increased oil consumption. Check oil level more frequently.
C	**5W-30**
D	**Synthetic 5W-30**
E	**Vanguard™ Synthetic 15W-50**

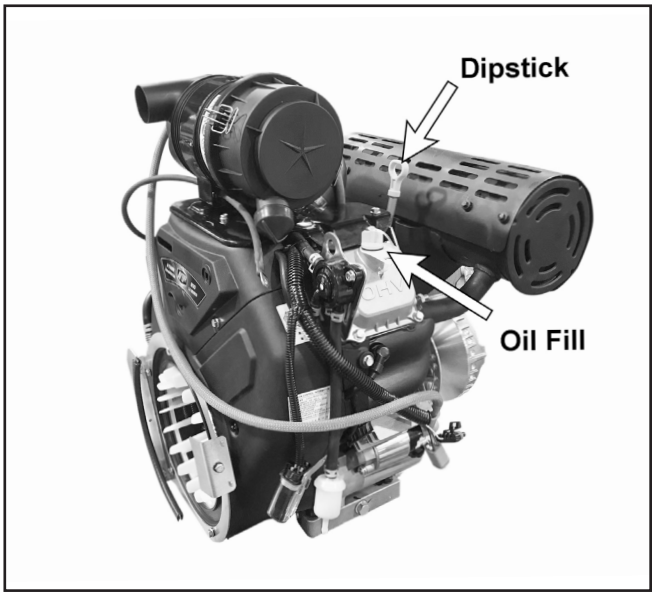
A - Engine ID B - Spark Plug C - Air Cleaner D - Dipstick E - Oil Fill
F - Oil Filter G - Oil Drain Plug

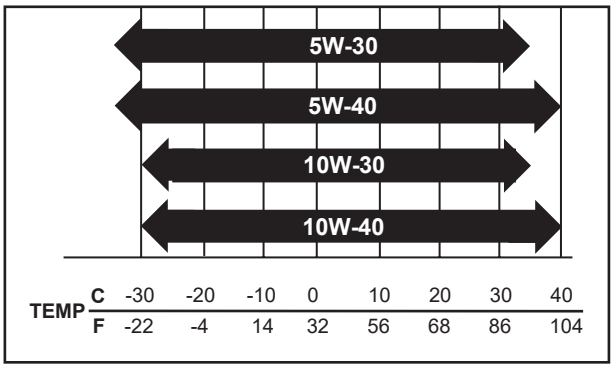
Briggs Vanguard 600/650/700 EFI - Oil Capcity 1.7 L 1.8 qts.)



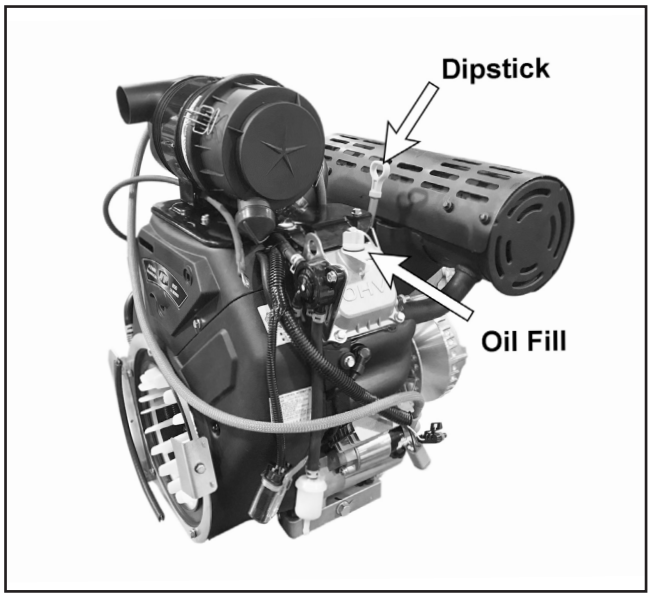


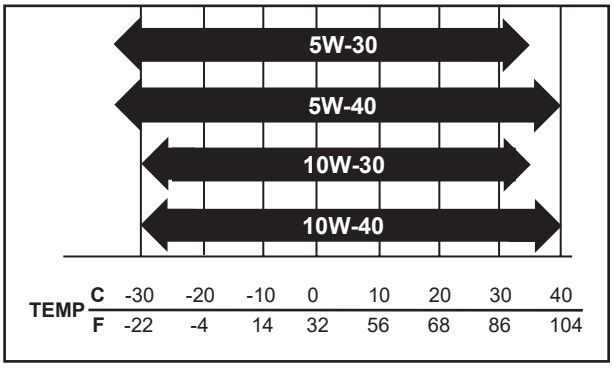
ODG 850 LX - With Filter: 2.4 qt. (2.3 L) - W/O Filter: 2.2 qt (2.1 L)





ODG 650 LX - With Filter: 1.9 qt.(1.8 L) - W/O Filter: 1.6 qt. (1.6 L)





Recommended Engine Oil

Use a high quality detergent oil of API (American Petroleum Institute) service class or ARGO Brand 5W40 synthetic oil, p/n 130-103. Choose the correct viscosity of oil for seasonal driving conditions. See chart oil charts for each engine.

Changing Engine Oil

During the initial engine break-in period, change the oil after the first 20 hours of operation. After the break-in period, change the engine oil every 50 operating hours, or more frequently if the vehicle is operated in dusty or dirty conditions.

Draining The Engine Oil

1. Start and warm up the engine so the oil will drain easily.
2. Level the vehicle so the oil will drain completely.

⚠ NOTE

There is limited space between the engine and power pack frame to place a collection tray at the oil drain plugs. Remove the oil from the engine by inserting a vacuum tube through the dipstick hole and using a vacuum pump such as the 638-02 TOPSIDER MVP VACUUM PUMP available through ARGO



Oil Filter Replacement

Change the oil filter every 100 hours during oil change. Before installing the new filter, lubricate the rubber filter gasket with fresh engine oil. Screw the filter on by hand until the gasket contacts the filter adapter. Tighten 1/2 to 3/4 turn more. Add the specified oil as shown in the recommended oil section of this service guide. Start and run engine to check for oil leaks. Stop engine and re-check oil level. Add oil if required.

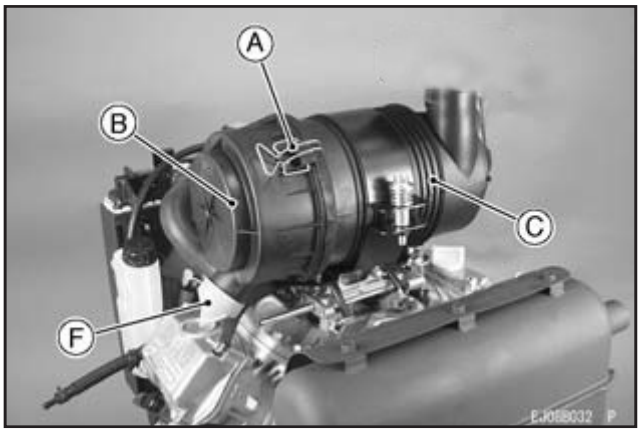
Refilling The Engine

Refill the engine through the oil fill port with the correct amount of oil referring to the chart in the recommended oils section. Make sure the appropriate grade of oil is used . As you add oil, frequently check the level with the dipstick. **DO NOT OVERFILL**. Start engine. Check for leaks. Stop the engine. Check the oil level. Add oil only to the "Full" mark on the dipstick.

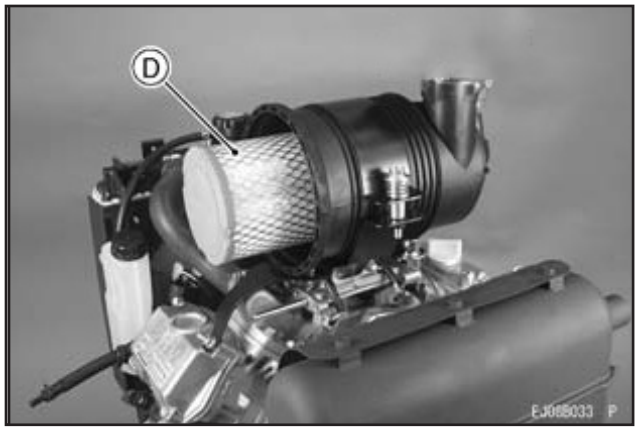
Air Filter (Kawasaki 800)

The Kawasaki FD851D engine is equipped with a heavy duty high density paper air cleaner element surrounding a canister style inner element. Cleaning is not recommended, each element should be replaced when dirty.

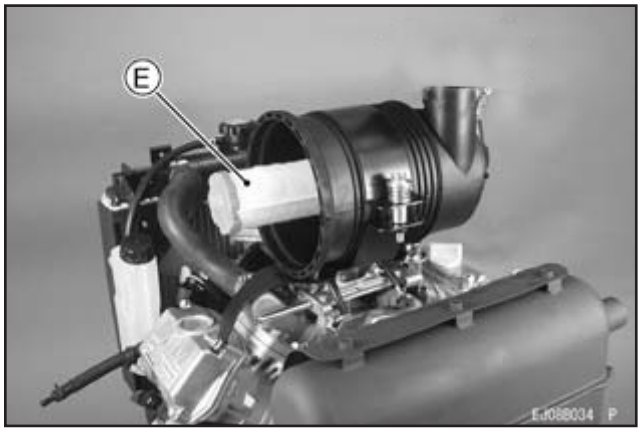
Check the paper air filter element every 100 hours of operation or more often under extremely dusty or dirty conditions.



A. Clamps **B. Air Cleaner Case**
C. Air Cleaner Body **F. Debris Collector Cap**



D. Primary Element



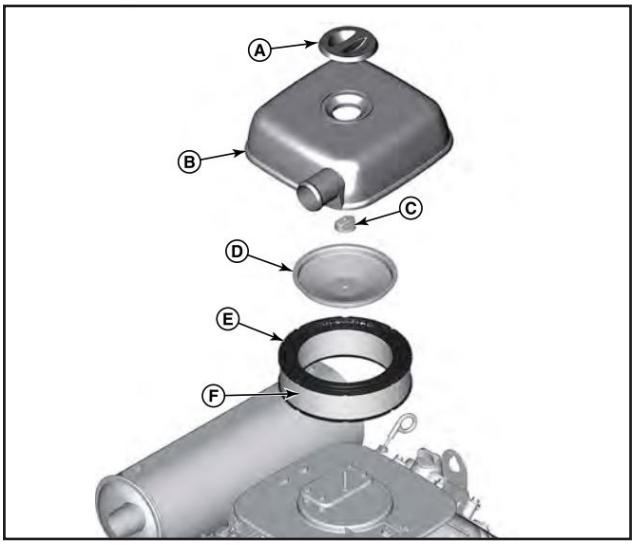
E. Secondary Element

Air Filter (Briggs Big Block 850/950)

Briggs Big Block EFI engines are equipped with a foam pre-cleaner and dry paper air filter element housed in an air cleaner assembly attached to the throttle body.
Replace or wash pre-cleaner in warm water with detergent. Rinse and allow to air dry. Saturate pre-cleaner with new engine oil; squeeze out excess oil. Replace the paper element.

Wash and oil the pre-cleaner after every 25 hours of operation or more often under extremely dusty or dirty conditions.

Check the paper air filter element every 100 hours of operation or more often under extremely dusty or dirty conditions.



A. Cover Fastener B. Cover C. Air Filter Element Retainer Fastener

D. Air Filter Element Retainer E. Air Filter Element F. Air Filter Foam Pre-Cleaner

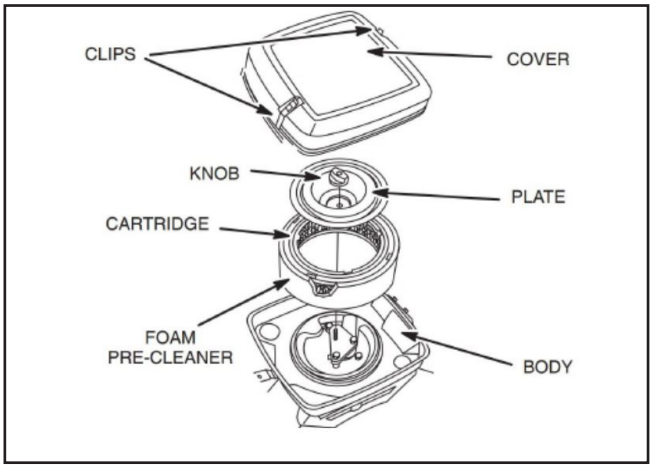
Air Filter (Briggs 600/650/700 EFI)

All Briggs and Stratton Vanguard engines are equipped with a foam pre-cleaner and dry paper air filter element housed in an air cleaner assembly attached to the throttle body/carburettor.

Replace or wash precleaner in warm water with detergent. Rinse and allow to air dry. Saturate precleaner with new engine oil; squeeze out excess oil. Replace the paper element.

Wash and oil the precleaner after every 25 hours of operation or more often under extremely dusty or dirty conditions.

Check the paper air filter element every 100 hours of operation or more often under extremely dusty or dirty conditions.



Air Filter (ODG 850/650)

ODG 850/650 engines are equipped with a heavy duty high density paper air cleaner element surrounding a canister style inner element. Cleaning is not recommended, each element should be replaced when dirty.

Check the paper air filter element every 100 hours of operation or more often under extremely dusty or dirty conditions.

